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Cloud IDS
with high performance using XDR

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CHAPTER 1
introduction

1.1Introduction:

In today's world, where reliance on cloud computing is continually increasing, organizations depend heavily on cloud infrastructure to store data and run applications and services. As dependence on the cloud grows, new challenges related to information security and sensitive data protection have emerged, with cyber-attacks evolving constantly to target cloud environments specifically. Consequently, there is a growing need for advanced and effective systems to monitor and detect threats and attacks in cloud environments quickly and reliably. This is where the Cloud Intrusion Detection System (Cloud IDS) project comes into play, offering a modern solution to address these challenges and enhance security levels within cloud computing environments.

1.2problem definition:

Cloud infrastructures and traditional Intrusion Detection Systems (IDS) suffer from a clear deficiency in detecting distributed and complex attacks targeting critical file transfer services (such as FTP). These systems often rely on isolated, node-specific monitoring, lacking both a holistic view and the ability to establish temporal correlations between events. The core limitation is as follows:

Traditional tools are unable to detect the granular processes and activities—stemming from file uploads and downloads—as they occur within the server environment Difficulty in detecting distributed, multi-stage attacks: The inability to correlate disparate alerts—such as distributed brute-force attempts, followed by the uploading of a malicious file with specific signatures, and culminating in data exfiltration—across multiple nodes and master controllers.

Slow immediate response: Reliance on human intervention or delays in correlating centralized logs, which allows an attacker to complete the stages of a breach before any automated preventive action can be taken to isolate the threat.

1.3Project goal:

The project aims to design and implement an advanced Cloud Intrusion Detection and Response System (Cloud IDS) with a distributed (multi-master/multi-node) architecture, specifically tailored to protect File Transfer Protocol (FTP) services. It provides the following capabilities:

Kernel-level monitoring: Integration of kernel-level tools (such as eBPF) to monitor FTP server activity and track file-related operations in real-time, while capturing essential data (file hash, size, and value).

Centralized analysis and log aggregation: A mechanism for aggregating and analyzing data across nodes and controllers to detect distributed attack patterns and reduce false

positives.
Advanced contextual correlation (XDR Correlation):
Leveraging XDR technologies to correlate outputs from network detection tools (such as Zeek) across successive stages of an attack—from reconnaissance and probing to unauthorized access and execution—thereby transforming disparate events into a cohesive attack scenario.
Immediate automated response: Automating security decision-making by issuing real-time commands to firewalls or Security Orchestration, Automation, and Response (SOAR) systems to isolate malicious IP addresses and block FTP sessions.

CHAPTER 2

Reference study

2.1 intro:

This section provides a theoretical overview of the relationship between cloud concepts (IaaS, PaaS, and SaaS), the principles of Internet Survey Definition Systems (ISS), and the use of the Falco system in a cloud environment. It also covers the concept of a secure cloud environment, its fundamentals, and its security. The discussion will explore how service model selection aligns with security and choice requirements, and how these elements impact system design, development costs, and operational expenses.

2.2 Cloud Definition

According to NIST's definition, the cloud is defined as a model for providing computing resources (such as servers, storage, and applications) on demand through technical interfaces, taking into account key characteristics: scalability, service-by-service management, and deployment flexibility (public, private, hybrid), in addition to specific security and compliance frameworks. [1]

Many organizations and researchers have defined the architecture for cloud computing. Basically the whole system can be divided into the core stack and the management. In the core stack, there are three layers: (1) Resource (2) Platform and (3) Application. The resource layer is the infrastructure layer which is composed of physical and virtualized computing, storage and networking resources.

The platform layer is the most complex part which could be divided into many sub-layers. a computing framework manages the transaction dispatching and/or task scheduling. A storage sub-layer provides unlimited storage and caching capability. The application server and other components support the same general application logic as before with either on-demand capability or flexible management, such that no

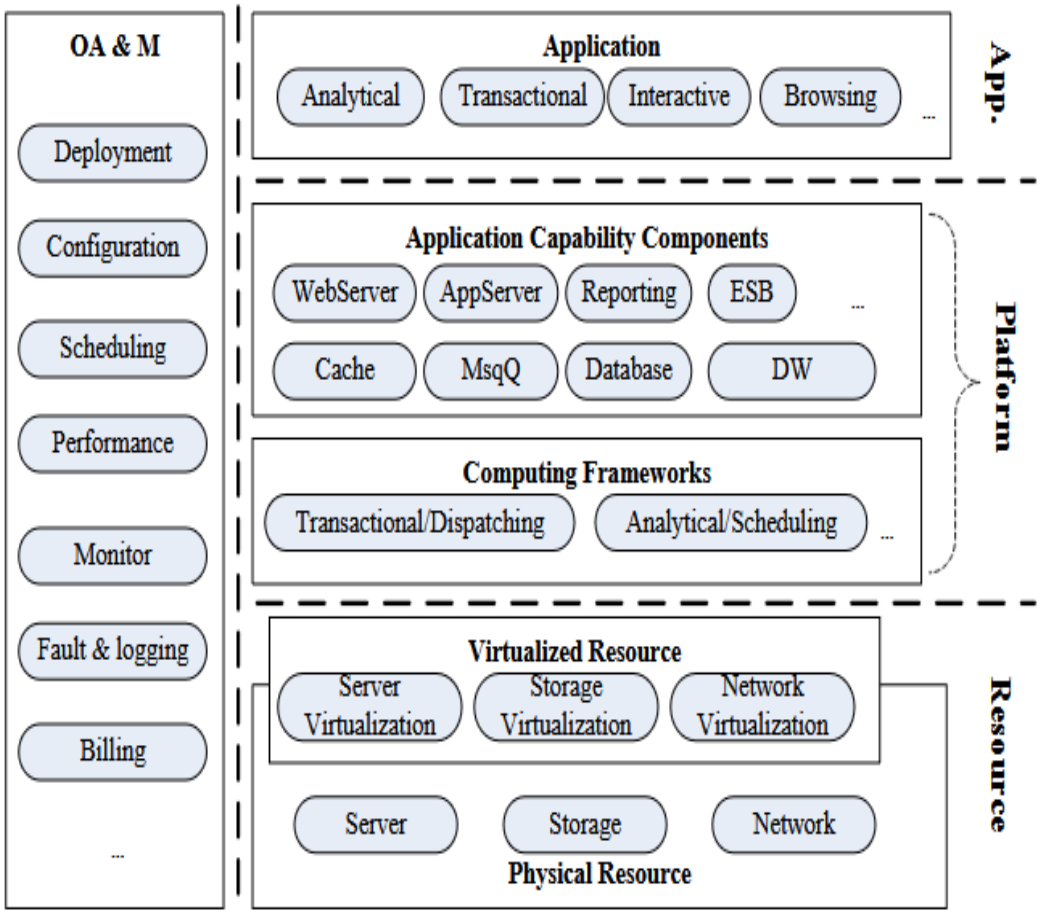
components will be the bottle neck of the whole system.This definition links the concept of services (IaaS/PaaS/SaaS)[1]

2.2.1 IaaS (Infrastructure as a Service): Providing a cloud-based infrastructure that defines a virtual location, storage, and network. It uses an infrared electronic controller. Examples: Amazon EC2, Microsoft Azure Virtual Machines, Google Compute Engine.[1]

2.2.2 PaaS (Platform as a Service): Providing an application development and operation platform with internet-based management and updates, allowing developers to focus on programming and integration. Examples: Google App Engine, Microsoft Azure Application Services, Heroku.[1]

2.2.3 SaaS (Software as a Service): Providing online applications without formal infrastructure for managing personal data or the platform. Examples: Google Workspace, Microsoft 365, Salesforce.[1]

Fig.1 The Reference architecture



2.3 Cloud Computing Categories:

There are diverse dimensions to classify cloud computing, two commonly used categories are: service boundary and service type.

From the service boundary's view, cloud computing can be classified as public cloud, private cloud and hybrid cloud. The public cloud refers to services provided to external parties. The enterprises build and operate private cloud for themselves. Hybrid cloud shares resources between public cloud and private cloud by a secure network. Virtual Private Cloud (VPC) services released by Google[8] and Amazon[9] are examples of Hybrid cloud.

* From the service type's view, cloud computing can be classified as Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS).

SaaS provide services to end users, while IaaS and PaaS provide services to ISV and developers - leaving a margin for 3-party application developers**.[1]**

2.4 Advantages and Risks:

The cloud computing is a Win-Win strategy for the service provider and the service consumer. We summarize the advantages as below:

2.4.1.Satisfy business requirements**:**

on demand by resizing the resource occu-

pied by application to fulfill the changing the customer requirements.

2.4.2.Lower cost and energy-saving:

By making use of low cost PC, customer-ized low power consuming hardware and server virtualization, both

CAPEX and OPEX are decreased.

2.4.3.Improve the efficiency of resource management:

through dynamic re-source scheduling

However there are also some major challenges to be studied.the risks:

2.4.4 Privacy and security:

Customer has concerns on their privacy and data security than traditional hosting service.

2.4.5 The continuity of service:

It refers to the factors that may negatively affected the continuity of cloud computing such as Internet problems, power cut-off, service disruption and system bugs. Followed are some typical cases of such problems: In November 2007, RackSpace, Amazon's competitor, stopped its service for 3 hours because of power cut-off at its data

center; in June 2008, Google App Engine service broke off for 6 hours due

to some bugs of storage system; In March 2009, Microsoft Azure experienced 22 hours' out of service caused by OS system update. Currently, the

public cloud provider based on virtualization, defines the reliability of ser-

vice as 99.9% in SLA.

2.4.6 Service migration:

Currently, no regularity organization have reached the agreement on the standardization of cloud computing's external interface.

As a result, once a customer started to use the service of a cloud computing

provider, he is most likely to be locked by the provider, which lay the customer in unfavorable conditions.

2.5 Kubernetes:

2.5.1 WHAT IS KUBERNETES?

Kubernetes is an open source container orchestration system. It manages containerized applications across multiple hosts for deploying, monitoring, and scaling containers. Originally created by Google, in March of 2016 it was donated to the Cloud Native Computing Foundation (CNCF).

Kubernetes, or "k8s" or "kube" for short, allows the user to declare the desired state of an application using concepts such as "deployments" and "services." For example, the user may specify that they want a deployment with three instances of a Tomcat

web application running. Kubernetes starts and then continuously

monitors containers, providing auto-restart, re-scheduling, and

replication to ensure the application stays in the desired state.[2]

2.5.2 KEY KUBERNETES CONCEPTS:

Kubernetes resources can be created directly on the command line but are usually specified using Yet Another Markup Language(YAML). The available resources and the fields for each resource may change with new Kubernetes versions, so it's important to double-check the API reference for your version to know what's available. It's also important to use the correct "apiVersion" that matches your version of Kubernetes. This Refcard uses the API from Kubernetes 1.12, released 27 September 2018.

POD:

A Pod is a group of one or more containers. Kubernetes will schedule all containers for a pod into the same host, with the same network namespace, so they all have the same IP address and can access each other using localhost.

2.5.3 Cluster Structure: Consists of control planes and worker nodes. Key Components: Pods (basic container units), Deployments (deployment and update management), Services (application access interface), ConfigMaps and Secrets (control of settings and sensitive data). Operation: kube-apiserver, etcd as a central database, kube-scheduler for node selection, controllers for managing desired state, kubelet on nodes, kube-proxy for network management. Deployment Management: Via YAML/JSON definitions and kubectl as a controller. Scaling and Recovery: Supports autoscaling, automatic reconfiguration, and load distribution via Services and Ingress.[3]

For Security and Compliance: Access Control: RBAC and resource permissions.

Network Policies: Network policies to restrict movement between containers.

Container Security: Pod Security Standards, image scanning, and security updates.

Secret Handling: Secrets and their encryption during transmission and storage.

Cryptography and Identity: TLS for components and necessary component authentication.

2.5.4 most famous versions of ks:[4]

2.5.4.1 K8s

The most comprehensive platform for managing and orchestrating application containers across multiple environments.

It orchestrates the automated deployment, scaling, updating, and maintenance of containers.

2.5.4.2 K9s

An interactive CLI for managing Kubernetes clusters more dynamically and easily from the command line.

it displays resources and allows for quick navigation between them, with fast commands for accessing and updating.

2.5.4.3 K3s

A lightweight Kubernetes distribution designed for resource-constrained environments like Edge and on-premises development.

It provides a complete Kubernetes experience with a simpler installation option and less complex configuration, using SQLite by default and a swappable database.[5]

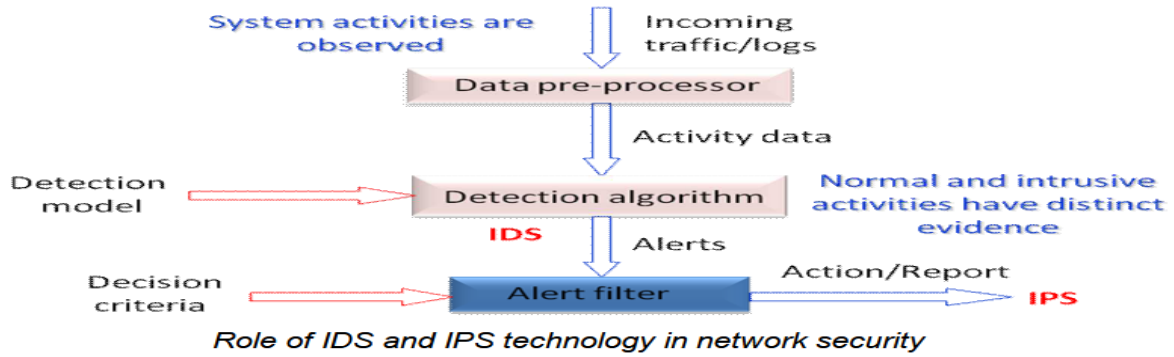
2.6 IDS:

IDS or Intrusion Detection Systems, are security components that aim to detect unusual and unauthorized activities within an organization's networks and systems and provide real-time alerts to enable rapid response. IDS typically operate in two main models: network-based IDS that monitors network traffic for known or suspected patterns, and host-based IDS that monitors system behavior and changes at the device level (files, processes, user permissions).

Detection mechanisms generally rely on two approaches: signature-based, which recognizes predefined attack patterns, and anomaly-based, which looks for behavior that deviates from the normal baseline. IDS operates within the organization's external protection layer as part of the cybersecurity stack including SIEM and SOC, and is used to generate alerts, analyze incidents, and document response. In cloud environments, examples include AWS GuardDuty, Azure Defender for Cloud, and Google Cloud IDS, which offer network- or host-based detection capabilities and integrate with reporting and investigation tools.

Key advantages include improved incident response, reduced recovery time, and enhanced security visibility, while facing limitations such as the need for careful policy configuration, alert-volume management, and potential gaps in advanced or complex attacks.[6]

Figure 2.How ids works :



Role of IDS and IPS technology in network security

2.6.1Types of IDS:

- Basic Types Network-Based IDS (NIDS): Monitors traffic across networks and interfaces.
- Host-Hosted IDS (HIDS): Monitors the activity of the system itself (files, logs, processes).
- Integrated IDS/IPS: Combines detection, response, and prevention capabilities into a single solution.

2.6.2 Mechanism of IDS:

- Monitoring: Compares real-time data with known safe models and attacks.
- Analysis: Uses signature-based prediction, machine learning, and anomaly-based behavior analysis.
- Alert and Response: Sends notifications and logs, and helps inform intrusion response policies.

Common examples: Network-based IDS (NIDS) and Hosted IDS (HIDS), as well as IDS/IPS systems integrated into some solutions.

2.6.3 Importance in Cloud:

Environments Protecting the access and data layer across SaaS/PaaS/IaaS services and applications. Detecting unauthorized activity across multiple pathways (virtual networks, containers, applications). Enhancing compliance and governance through security logging and rapid remediation solutions. Adapting to evolving threats and ensuring operational flexibility across multi-region deployments.

2.7 ZEEK

2.7.1 What is ZEEK:

Zeek (formerly known as **Bro**) is a free, open-source network analysis framework and network security monitor (NSM). Created by Vern Paxson in 1995 at the Lawrence Berkeley National Laboratory (LBNL) and renamed in 2018, it passively observes network traffic without active intervention. [7]

Core Capabilities & Functionality

• **Protocol & Traffic Analysis:** Inspects deep application-layer protocols such as HTTP, DNS, SSL/TLS, and SSH.

• **Structured Telemetry Logs:** Converts raw network packets into comprehensive, high-fidelity log files (such as conn.log for connections, http.log for web traffic, and files.log for transferred files) formatted as TSV or JSON.

• **Unique Connection IDs (UIDs):** Automatically assigns a unique UID to each flow, allowing analysts to link separate records together to reconstruct full attack timelines.

• **Scriptable Engine:** Uses a Turing-complete, domain-specific programming language that lets security teams customize detection logic and policy rules.

• **Threat Intelligence & Forensics:** Supports real-time matching against threat indicators and integrates smoothly with SIEM and analytical pipelines

2.7.2 How it works :

This architecture relies on the distribution of roles and tasks among various types of nodes, which are interconnected via a messaging layer based on a publish/subscribe system:

1. Zeek Cluster Nodes and Their Roles

• Worker Nodes:
• These nodes handle the bulk of the workload and the core processing tasks; they are responsible for sniffing network traffic, deconstructing data packets, and reassembling network flows.
• They analyze network protocols and generate security-related and operational "events."
• They do not write logs locally; instead, they forward data to higher-level nodes.

• Manager Node:
• This serves as the "brain" and central node that oversees the entire cluster.
• It receives events that require a comprehensive, network-wide perspective and executes centralized policies and scripts.
.

2.8 File browser :

2.8.1 what is file browser :

File Browser, also known as File Explorer, is a user interface tool that lets users browse and organize files and folders stored on a computer or within a cloud computing environment. It can operate as a graphical or text-based interface that enables sorting, searching, opening, moving, copying, and deleting files, while providing detailed information such as access permissions, file sizes, and creation/modification dates. Types of File Browser vary by environment; desktop browsers are common in operating systems like Windows, macOS, and Linux, in addition to built-in browsers within servers' file management systems and web applications, and sometimes it integrates with cloud functions such as displaying storage content from AWS S3 or Google Cloud Storage. The experience relies on a usability-focused design that facilitates navigation of the hierarchical folder tree, with immediate options for search, filtering, and multi-select, as well as file management capabilities like version awareness, permission control, and integration with collaborative tools. In security and governance environments, a File Browser can include fine-grained access controls, secure transfer protocols, user activity tracking, and possibly certificate embedding and data encryption settings to protect content during browsing and transfer. If you'd like a specialized clarification (e.g., Linux GUI browsers or cloud-based browsers) or a comparison among several solutions

2.8.2 How it works:

Background storage: Relies on a cloud provider that stores files as objects with specific paths and access links. Viewing interface: Provides a graphical interface similar to a file explorer for browsing and interacting with folders and files. Permissions and sharing: Supports access permissions, link sharing, and access control mechanisms. Basic operations: Upload, download, create folders, delete, rename, search, and move.

2.8.3 Popular Features:

Easily view and organize files using simple folder structures. Drag and drop functionality, and batch uploads. Access control and encryption during transfer and storage. Quick search and filtering by type, size, and date. Integration with other cloud tools (SIEM, CI/CD, backup systems).

2.8.4 The security service that provides a cloud file browser (cloud file browser):

1-Access and Identity Control User authentication on sessions Precise deployment across roles and permissions on resources Multi-Factor Authentication (MFA) support .

2-Data Encryption and Protection Teleread-in-Transit (TLS) Teleread-in-Atom (AES/Other depending on the provider) Key management via KMS services.

3-Trace and Persistence Log of Japanese user actions (audit logs) Access, change, and download reports Support for core standards (e.g., GDPR, HIPAA depending on context)

4-Tracking and Compliance User activity and process logging (Audit logs) Access, change, and download reports Compliance standards support (e.g., GDPR, HIPAA as contextual)

5-Link and Temporary Access Management Pre-signed URLs for sharing control Modified and trackable sharing permissions 6-Security and Configuration Policies Bucket/Object policies Compliance checks and configuration checks Alerts for unusual activity or unauthorized access.

7-Tamper and Data Loss Protection Versioning support for restoring previous versions Protection against accidental deletion (Object lock/immutability on some providers) 8-Network Integrity and Security Isolating networks and applications between S3/GCS/Azure environments Access control via network standards (VPC endpoints, firewall rules)

2.9 XDR :

(**Extended Detection and Response**) is a unified cybersecurity architecture that collects, normalizes, and correlates threat data across multiple security layers—including endpoints, networks, cloud workloads, email systems, and user identities—into a single platform.[8]

2.9.1 HOW IT WORKS:

Data Ingestion: Gathers raw telemetry and logs from across the entire IT environment instead of isolated silos.

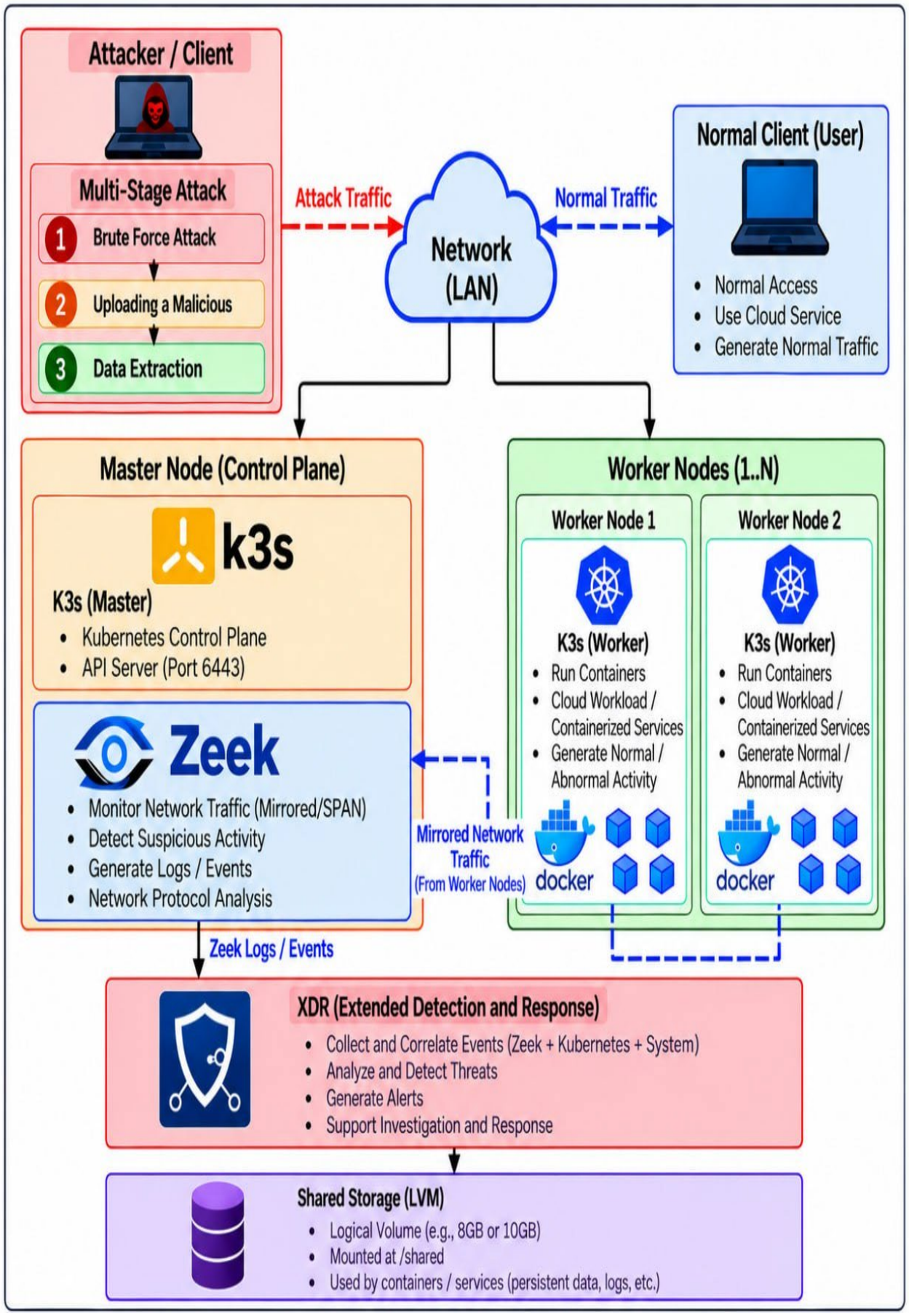
• **AI & Correlation:** Uses artificial intelligence, machine learning, and behavioral analytics to connect related events and trace complex, multi-stage attacks.

• **Automated Response:** Executes automated remediation actions—such as isolating a compromised host, blocking malicious network traffic, or disabling an account.

2.10 Conclusion:

This section covers several important terms, including: basic definitions of cloud and its IaaS/PaaS/SaaS services; deployment models; Kubernetes fundamentals with examples of the K9s tool and the K3s option; Falco and IDS systems; and security in cloud environments. It also discusses potential examples and comparisons, as well as source options for references and documentation in addition of file browser and how it works

PROPOSED SYSTEM


Attacker / Client
Multi-Stage Attack
1 Brute Force Attack
2 Uploading a Malicious
3 Data Extraction
Attack Traffic
Network (LAN)
Normal Traffic
Normal Client (User)
• Normal Access
• Use Cloud Service
• Generate Normal Traffic
Master Node (Control Plane)
k3s
K3s (Master)
• Kubernetes Control Plane
• API Server (Port 6443)
Zeek
• Monitor Network Traffic (Mirrored/SPAN)
• Detect Suspicious Activity
• Generate Logs / Events
• Network Protocol Analysis
Worker Nodes (1..N)
Worker Node 1
K3s (Worker)
• Run Containers
• Cloud Workload / Containerized Services
• Generate Normal / Abnormal Activity
docker
Worker Node 2
K3s (Worker)
• Run Containers
• Cloud Workload / Containerized Services
• Generate Normal / Abnormal Activity
docker
Mirrored Network Traffic (From Worker Nodes)
Zeek Logs / Events
XDR (Extended Detection and Response)
• Collect and Correlate Events (Zeek + Kubernetes + System)
• Analyze and Detect Threats
• Generate Alerts
• Support Investigation and Response
Shared Storage (LVM)
• Logical Volume (e.g., 8GB or 10GB)
• Mounted at /shared
• Used by containers / services (persistent data, logs, etc.)

RESEARCH STUDIES FOR CLOUD INTRUSION DETECTION SYSTEM (CLOUD IDS)

Literature Review Summary

Title	journal	Pub year	achievement	benefit
EIDS: A Cloud Intrusion Detection System with High Performance and Maintainability	ACM Transactions on Computer Systems	2026	• Introduces an EIDS framework designed for high performance and maintainability. • Divides data into two sections and develops a micro-VM scheduler.	Provides a practical high-performance architecture relevant to Cloud IDS, particularly for reducing latency and improving throughput and maintainability.
Intrusion Detection Systems in Cloud Computing Paradigm: Analysis and Overview	Complexity	2022	• Reviews IDS approaches in cloud-computing environments. • Compares FCM-ANN, SVM-ANN, FCM-SVM, and SMO-ANN. • Uses NSL-KDD and UNSW-NB15 datasets and common evaluation metrics.	Supports understanding of cloud IDS approaches and limitations, and informs the selection of machine-learning techniques and evaluation metrics
Towards an Intelligent Intrusion Detection System to Detect Malicious Activities in Cloud Computing	Journal of Computer Virology and Hacking Techniques	2023	• Proposes a hybrid IDS model for cloud computing. • Focuses on accuracy, feature reduction, data imbalance, and computational efficiency.	Offers ideas for improving detection accuracy and efficiency, and for addressing class imbalance in a cloud-based intrusion detection system.
Adversarial Threats to Cloud IDS: Robust Defence With Adversarial Training and Feature Selection	IEEE Access	2025	• Evaluates adversarial attacks including FGSM, PGD, and DeepFool. • Reports that such attacks can reduce IDS detection accuracy; adversarial protection improves robustness.	Highlights the need to test the project's detector against adversarial evasion and consider robust training and feature selection.

Utilizing IDS and IPS to Improve Cybersecurity Monitoring Process	Journal of Cyber Security and Risk Auditing	2025	• Discusses the differences between IDS and IPS, their detection methods, and the value of integrating them.	Helps explain the role of IDS in the project and provides background for possible future integration of prevention or response capabilities.
Building a Cloud-IDS by Hybrid Bio Inspired Feature Selection Algorithms Along With Random Forest Model	IEEE Access	2024	• Develops a Cloud IDS using hybrid feature selection (GA + GOA) and a Random Forest model to improve attack detection	Provides a reference for improving feature selection and applying Random Forest to enhance attack detection in the Cloud IDS project.
SOAR4DER: Security Orchestration, Automation, and Response for Distributed Energy Resources	Power Systems Cybersecurity (Springer book chapter	2023	• Proposes SOAR4DER, combining multiple IDS data streams with physical and cyber features. • Uses SOAR playbooks to automate defence and restore systems to a known-good state. • Laboratory tests reported detection and response times under 30 seconds for the tested attack scenarios	Offers a model for future automated response in Cloud IDS, such as triggering playbooks from IDS alerts and safely containing attacks. The original study focuses on distributed energy resources.

An Overview of EDR Serviceability for Security Information and Event Management (SIEM)	Journal of Information Assurance and Security, 21(2), 52–88	2026	• Reviews Endpoint Detection and Response (EDR) and its integration with SIEM. • Discusses real-time telemetry, behavioural analytics, AI/ML, log correlation, open-source tools, scalability, and automated response	Supports future integration of host-level telemetry and centralized log analysis with Cloud IDS, and highlights operational considerations such as interoperability, scalability, and response automation.
AI-Driven Threat Detection and Automated Incident Response for Enhancing Network Security	Network, 6(2), Article 32 (MDPI)	2026	• Presents an integrated framework combining signature-based, anomaly-based, and ML/DL detection with automated incident response. • Evaluates multiple datasets and reports 96–97% detection accuracy; mean response time decreases from about 45 minutes to under 30 seconds.	Provides a directly relevant architecture for extending Cloud IDS with hybrid detection, machine-learning analysis, centralized monitoring, and automated containment. Results are those reported by the paper and would need separate validation in our environment.

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